



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Mechanical engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: Heating ventilation and
air conditioning

Course Code: PEC-ME415

Date: 19/08/2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Solve any 5 questions.

Q.N.		Marks	CO	BT
Q.1	Explain the working of a vapour compression refrigeration system with a neat sketch.	4	CO2	BT1,2
Q.2	Compare Window AC and Split AC systems in terms of structure, function, and usage.	4	CO1	BT2
Q.3	Explain the components and working of a Centralized Air Conditioning system.	4	CO1	BT1,2
Q.4	Describe the applications of HVAC systems in industrial and commercial buildings.	4	CO1	BT1
Q.5	Classify HVAC systems and explain any two types with suitable applications.	4	CO1	BT2
Q.6	Write a note on any two air distribution systems used in HVAC. (i)CAV (Constant Air Volume), (ii)VAV (Variable Air Volume).	4	CO2	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply the concepts of psychometrics and thermodynamics to heating and cooling analysis.
CO 2	Use basic concepts from heat transfer to determine heat gained or lost from a building.
CO 3	Design air-handling systems using concepts from fluid dynamics.
CO 4	Apply good engineering principles to meet the requirements for air quality control and comfort conditions.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE I (Semester-1) 2025-26

En.Y. B. Tech.

Mechanical Engineering

Finite Element Analysis

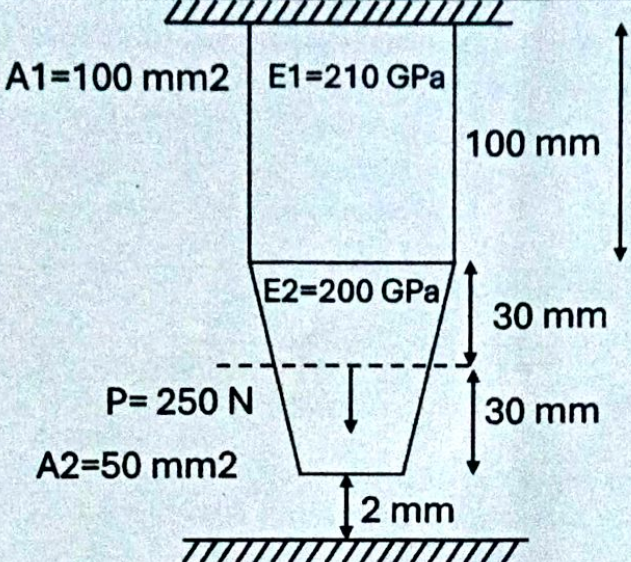
PEC-ME403

Date: 19-Aug-25

Time: 11:00 To 11:45

Max. Marks: 20

Note: All questions are compulsory

Q.N.	Question	Marks	CO	BT
Q.1	Discuss mesh Convergence study	2	CO1	BT1
Q.2	Discuss various stages of FEM for ANSYS software	6	CO3	BT2
Q.3	 Find out nodal displacement, stresses and reaction.	12	CO2	BT3

Course Outcomes (CO)

CO 1 Analyze and design real world components

CO 2 Suggest whether the given solid is safe for the load applied.

CO 3 Select proper methodology for designing and solving a design problem.

CO 4 Work with different software skills sets required for practical problem.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

**SGGS Institute of Engineering and Technology, Vishnupuri, Nanded****ISE-I (Semester-I) 2025-26****Final. Year. B. Tech.****Mechanical Engineering Department****Renewable Energy****PEC-ME414****Date: 18/08/25****Time: 11:00 to 11:45 am****Max. Marks :20**

- Note:**
- All questions are compulsory.
 - Assume suitable data if necessary.
 - Use of non-programmable calculator is permitted.
 - This paper contains ONE page.

Q.N.	Question	Marks	CO	BT
Q.1	A wind at standard 1 atm pressure and 30 °C has a velocity of 12 m/s, the turbine diameter is 100 m is operating at a speed of 50 rpm at maximum efficiency. Assume power coefficient as 0.35. Calculate: 1. Total power available in the wind 2. Toal power density in the wind system 3. Obtainable reasonable power 4. Obtainable reasonable power density 5. Torgue at maximum efficiency 6. Tip Speed ratio of wind turbine	06	CO3	4
Q.2	What do you mean by Renewable Energy? What are the different forms of Renewable Energy	04	CO1	1
Q.3	Discuss Environmental aspects of electric energy generation	04	CO1	2
Q-4	Derive the equation of Betz Limit in wind turbine	06	CO3	3

Course Outcomes (CO)

- CO 1: Interpret energy reserves of India and potential of different energy sources.
CO 2: Measure the solar radiation parameters and performance of different solar collectors.
CO 3: Calculate different parameters of wind turbine rotor.
CO 4: Appraise the importance and applications of geothermal and ocean energy.
CO 5: Demonstrate knowledge in field of fuel cells and potential for power generation

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Class: Final Year, B.Tech.

Subject: Operations Research

Subject Code: HMC-ME402

Date: 21/08/2025

Time: 11.00-11.45

Max. Marks: 20

NB:

- I. This paper contains Two pages (**Three Questions**). Figures to right indicate full marks. All Questions are Compulsory.
- II. **Outline your solution in clear, sequential steps.** Assume suitable data wherever necessary and clearly state it.

Q.No.

Questions

Marks

CO

BT

1. The following table shows all the necessary information on the available supply from each warehouse, the requirement of each market and the unit transportation cost in Thousands of Rupees from each warehouse to each market.

		<u>Markets</u>				
		A	B	C	D	
<u>Warehouses</u>	I	6	3	5	4	22
	II	5	9	2	7	15
	III	5	7	8	6	8
		7	12	17	9	

The manager of the organisation has worked out the following schedule from experience: 12 units from I to B, 1 unit from I to C, 9 units from I to D, 15 units from II to C, 7 units from III to A and 1 unit from III to C.

- a) Check if the manager has made the optimal schedule. If not Optimal, find the optimal schedule and minimum total shipping cost. (07 Marks)
- b) If the manager is approached by a carrier of route III to B who offers to reduce his rate in the hope of getting some business, by how much must the rate be reduced by the driver before the clerk should consider giving him business? (01 Mark)

2. Use the graphical method to solve the following LP problem –

07 CO1, BT3
2

$$\text{Max. } Z = (-x_1 + 2x_2)$$

$$\text{Sub. To, } x_1 - x_2 \leq -1,$$

$$-0.5x_1 + x_2 \leq 2,$$

$$x_1, x_2 \geq 0.$$

Clearly highlight the *common feasible region*. And comment on the nature of the solution (*Unique optimal / Multi-optimal / Degenerate / Infeasible*) with justification.

3. A Firm plans to purchase at least 200 quintals of scrap containing high quality metal X and low-quality metal Y. Scrap can be purchased from two suppliers A and B. Scrap must contain 100 quintals of metal X and no more than 35 quintals of metal Y. The percentage of X and Y metals in terms of weight in the scrap supplied by A and B are given below –

05 CO4 BT3

Metal	Supplier A	Supplier B
X	25%	75%
Y	10%	20%

The price of A's Scrap is \$200 per quintal and that of B's is \$400 per quintal. Formulate this problem as an *LP model*, to determine the quantity to be purchased from each supplier so that the cost of acquisition is minimum.
